BRAZILIAN E-COMMERCE PUBLIC DATASET BY OLIST

This project is based on the **Brazilian E-Commerce (Olist) Dataset** , which contains real – world transactional data of multiple online marketplace in Brazil. It includes information of more than 1,00,000 orders from 2016 to 2018.

The dataset includes multiple tables such as **Customers , Orders , Ordered items , Products and Payments** which provides a multi-dimensional view of each order, covering aspects such as order status, price, payment, freight performance, customer location, product attributes, and customer reviews.

Key Components of the Dataset :

1. **Orders Data :** This contains order details such as Order ID, Customer ID ,Purchase timestamp, Order Status , Expected Delivery Date etc.
2. **Customers Data :** This contains information about buyers , including their city and state.
3. **Order Item Data :** The order item table contains information about the items in each order, including product ID, seller ID, shipping limit date, and price.
4. **Product Data :** This table includes details about products, including product ID, product name length and product categories , which help in analysing sales across different segments.
5. **Payment Data :** Contains transaction details like payment type and payment value.

SQL IMPLEMENTATION

❖ STEP 1 : DATABASE CREATION

Objective :

To create and use that database for storing and analysing the e-commerce data.

SQL Query :

```
CREATE DATABASE E_COMMERCE;
USE E_COMMERCE;
```

Output :

Database was successfully created and set as the working environment.

INSIGHTS :

This step ensures that all the tables and operations are organized within a specified database for efficient data management.

❖ STEP 2 : IMPORT THE DATA USING FLAT FILES

Objective :

To import raw ecommerce data from csv into SQL Server for further processing and analysis of the data.

Process :

The dataset files like (Customers , Orders , Order Items , Products and Payments) were imported into SQL server using IMPORT FLAT FILE option in SSMS (SQL Server Management Studio).

Each Table was loaded and SQL Server automatically created the table structure based on the data.

Output :

All 5 Tables were successfully imported as individual tables :
OLIST_Customers
OLIST_Orders
OLIST_Ordered_Items
OLIST_Products
OLIST_Payments

INSIGHTS :
Importing Data through flat files provides a easy , quick and efficient way to load the raw datasets into SQL Server without manually creating table structure or schemas. It also ensures that the data is ready for querying and transformation.

❖ STEP 3 : DATA PREVIEW

Objective :

To understand the structure and content of the tables that we have imported in order to understand the dataset before performing any analysis or transfomrations.

SQL Query :

```
--Preview Customers Table
SELECT Top 10 *
FROM OLIST_Customer;

--Preview Orders Table
SELECT Top 10 *
FROM OLIST_Orders;

--Preview Ordered Items Table
```

```
SELECT Top 10 *
FROM OLIST_Ordered_Items;

--Preview Products Table
SELECT Top 10 *
FROM OLIST_Products;

--Preview Payments Table
SELECT Top 10 *
FROM OLIST_Orders_Payment;
```

Output :

The First 10 records of each table were displayed , providing an overview of columns , datatypes and values.

INSIGHT :

This step helped me in understanding how the data is distributed across multiple tables and also in identifying key columns such as :
Customer_id (that links customers and orders table together),
Order_id (link across orders , items , and payments),
Product_id (link between products and order items) etc.

❖ STEP 4 : DATA VALIDATION AND RELATIONSHIP CHECK

Objective :

To validate the relationships between different tables and ensures that the data can be joined using common keys.

SQL Query :

```
   -- Joining Orders and Customers Table together to understand the
relationship:
Select
    o.order_id
    ,o.customer_id
    ,c.customer_city
    ,c.customer_state
    ,o.order_purchase_timestamp
From
    Olist_Orders as o
Join
    Olist_Customer as c
ON
   o.customer_id = c.customer_id;
```

```
--Adding Order Items Table in Customers and Orders Table
Select
     o.order_id
     ,c.customer_city
     ,oi.product_id
     ,oi.price
From
    Olist_Orders as o
Join
    Olist_Customer as c
On
   o.customer_id = c.customer_id
Join
   Olist_Ordered_Items as oi
ON
   o.order_id = oi.order_id
 ;
```

Output :

The results ensures that :

- Each order is linked to a customer
- Each order can have multiple products.

INSIGHTS:

This step helped in identifying the relationship between tables, such as :

One customer can have multiple orders.

One order can contain multiple items.

One order can have multiple payment method.

❖ STEP 5 : DATA INTEGRATION

Objective :

To combine multiple tables into a single consolidated dataset for easier analysis and reporting.

SQL Query :

```
   --Combining all the tables into 1 final Table
SELECT
    o.order_id
    ,o.customer_id
    ,c.customer_city
    ,c.customer_state
    ,o.order_purchase_timestamp
    ,o.order_status
    ,oi.product_id
    ,oi.price
    ,oi.freight_value
    ,p.payment_type
    ,p.payment_value
    ,pr.product_category_name
INTO
    Final_dataset_org
FROM
```

```
    Olist_Orders as  o
JOIN
    Olist_Customer as c
ON
    o.customer_id = c.customer_id
JOIN
    Olist_Ordered_Items as oi
ON
    o.order_id = oi.order_id
JOIN
    Olist_Orders_Payment as p
ON
    o.order_id = p.order_id
JOIN
    Olist_Products as pr
ON
    oi.product_id = pr.product_id
 ;
```

Output :

A new table called Final_dataset_org was created containing combined information from all the tables such as customers , orders , order items , products and payment.

INSIGHTS:

Since the data was originally stored in multiple tables , performing joins helped in creating a centralized dataset.

This step made analysis easier by bringing together :

- Customer Information
- Order Details
- Product Data
- Payment Transaction.

It also highlights real world relationships like :

One customer = multiple orders

One order = multiple products

One order = multiple payments

❖ STEP 6: DATA CLEANING

Objective:

To improve the quality of data by handling missing values if any , removing dupliactes , correcting unusual data formats and removing irrelevant data.

SQL Query :

```sql
--Copy of Original
Select *
Into
   Cleaned_Dataset
From
      Final_Dataset_org;
```

INSIGHTS:

A separate working table to perform cleaning operations without changing the original data.

```sql
--Checking For Null Values:
Select
     Count(*)
     As Null_Payment
From
    Cleaned_Dataset
Where
    payment_value
    Is Null;

--Checking For Duplicates
Select
      order_id
      ,Count(*)
      As 'Dupliacte Orders'
From
    Cleaned_Dataset
Group By
     order_id
Having
     Count(*) > 1;


Select
      customer_id
      ,Count(*)
      As 'Dupliacte Customers'
From
    Cleaned_Dataset
Group By
     customer_id
Having
     Count(*) > 1;
```

INSIGHTS :

These duplicates may exists due to joins.

```sql
Select
     Distinct *
Into
    Cleaned_Data_distinct
From
    Cleaned_Dataset
;


--Fixing Date Format
Alter Table Cleaned_Data_distinct
Add order_date Date;
```

```sql
Update Cleaned_Data_distinct
SET order_date = Cast(order_purchase_timestamp
As Date)


--Dealing with Negatives
Delete From Cleaned_Data_distinct
Where payment_value <= 0;


--Handle Missing Category
Update Cleaned_Data_distinct
SET product_category_name = 'Unknown'
Where product_category_name Is Null;
```

❖ STEP 7: DATA ANALYSIS & INSIGHTS

Objective :

To analyze the cleaned dataset and extract meaningful insights related to revenue , customer behavior, and sales trend .

SQL Query :

```sql
--What is total revenue and total number of Orders?
Select
     Count(Distinct order_id)
     As Total_Orders
     ,Round(Sum(payment_value),2)
     As Total_Revenue
From
       Cleaned_Data_distinct;
```

INSIGHT:

This ecommerce business generated a total revenue of **1.68 crore** from **98,665 orders** , this shows that company has a good number of customer and steady sales.

I

```sql
--What is Monthly Sales Trend ?
Select
     Format(order_date, 'yyyy-MM')
     As 'Month'
     ,Round(Sum(Payment_value),2)
     As 'Revenue'
From
    Cleaned_Data_distinct
Group By
     Format(order_date, 'yyyy-MM')
Order By
    'Month';
```

INSIGHT:

Sales were very low in the beginning , especially in 2016 , but they started increasing steadily from **2017 onwards** . The **highest sales** were seen around **late 2017 and early 2018**, where revenue crossed **12 lakh** in some months.

This shows that the business is growing overtime and gaining more and more customers. These very low values in 2016 and 2018-09 may be because of incomplete data , not actual low sales.

```
--What are Top 10 Product Categories ?
Select
    Top 10
    product_category_name
    ,Round(Sum(payment_value),2)
    As Revenue
From
    Cleaned_Data_distinct
Group By
    product_category_name
Order By
    Revenue Desc;
```

INSIGHTS :

The highest sales or revenue is generated from **beauty & health (beleza_saude)** with around **14.9 lakh** , followed closely by **bed , bath & home (cama_mesa_banho)** with **14.5 lakh .**

Other strong categories include watches & gift and sports & leisure , each contributing more than 12 lakh.

This shows that customers are spending more on **lifestyle and daily-use-products** , making these categories the key drivers of revenue.

```
--What are Top States by Revenue ?
Select
    customer_state
    ,Round(Sum(payment_value),2)
    As 'Revenue'
From
    Cleaned_Data_distinct
Group By
    customer_state
Order By
    Revenue Desc;
```

INSIGHTS :

The State "**SP**"(Sau Paulo) generated the highest revenue around 63.1 lakh , which is much higher than the other states .

It is followed by “RJ” with 22.5 lakh and “MG” with 19.6 lakh.

This shows that most of the sales are concentrated in a few major states , especially “SP” , which is the main revenue generator.

```
--What are the most prefered payment methods ?
Select
     payment_type
     ,Count(*) As
     'Transaction'
     ,Round(Sum(payment_value),2)
     As Revenue
From
    Cleaned_Data_distinct
Group By
    payment_type
Order By
    Revenue Desc;
```

INSIGHTS :

The most preferred payment method is “**CREDIT CARD**”, with around **79,258 transactions** generating **1.31 crore revenue**. The second most used method is **boleto**, with 20,288 transactions and around 30.3 lakh revenue.

Other methods like voucher and debit card are used much less , contributing only a small part of total transactions.

Business can ensure smooth transaction and maybe give offers on cards to increase sales.

```
--What is customers purchasing power ?
Select
     Round(Sum(payment_value)
     / Count(Distinct order_id),2)
     As avg_order_value
From
    Cleaned_Data_distinct;
```

INSIGHTS :

The average order value is **170.41** , which shows that customer *spending behavior* or their *purchasing power* is **moderate amount per order** .

This means that most of purchases are small to medium in value , and there is an opportunity to increase the sales by encouraging the customers to buy more in one order by giving them offers like combos , discounts or vouchers.

```
--Who are the Top 5 Customers ?
Select
     Top 5
     customer_id
     ,Round(Sum(payment_value),2)
```

```sql
     As Total_spent
From
    Cleaned_Data_distinct
Group By
    customer_id
Order By
    total_spent Desc;
```

INSIGHTS :

The top customer spent around 16k which is very small compared to total revenue of 1.68 crore. Even the top 5 customers together contributing a very small portion of total sales.

This shows that the business is not depend on few customers, but instead earns revenue from a large number of regular customers, which reduces the dependency on a few high value customers and makes the business more stable.

```sql
--Which days have the highest order volume?
Select
     Datename(Weekday , order_date)
     As 'day'
     ,Count(Distinct order_id) As
     'Orders'
From
    Cleaned_Data_distinct
Group By
    Datename(Weekday , order_date)
;
```

INSIGHTS :

The highest number of orders were placed on **MONDAY (16068)** , followed by **Tuesday (15,831)** and **Wednesday (15,425).** This shows that customer prefer shopping at the start of the week.

On the other hand , Saturday (10,813) and Sunday (11,888) have comparatively lower orders , indicating reduced activity during weekends.

So business can plan promotions or offers during weekends to increase sales.